



### **Semester 2 Examinations 2017/2018**

|                             |                                                                 |
|-----------------------------|-----------------------------------------------------------------|
| <b>Exam Code(s)</b>         | 3BMS2, 3BPT1, 3BS9, 3BPT1, 3FM2, 4BME1, 4BMS2, 4BS2             |
| <b>Exam(s)</b>              | 3 <sup>rd</sup> Arts, 3 <sup>rd</sup> & 4 <sup>th</sup> Science |
| <b>Module Code(s)</b>       | MP307                                                           |
| <b>Module(s)</b>            | Modelling II, Mathematical Modelling                            |
| <b>Paper No.</b>            | 1                                                               |
| <b>Repeat Paper</b>         | No                                                              |
| <b>External Examiner(s)</b> | Professor Y. Fu                                                 |
| <b>Internal Examiner(s)</b> | Dr P. Piironen<br>Professor M. Tuite                            |

**Instructions:** Marks are awarded for question 1 in Section A **and** the TWO best questions in Section B.  
All questions carry the same weight.

|                               |                     |
|-------------------------------|---------------------|
| <b>Duration</b>               | 2 hours             |
| <b>No. of Pages</b>           | 6                   |
| <b>Discipline(s)</b>          | Applied Mathematics |
| <b>Course Co-ordinator(s)</b> | Dr P. Piironen      |

**Requirements:**

|                         |            |                                     |    |                          |
|-------------------------|------------|-------------------------------------|----|--------------------------|
| Release in Exam Venue   | Yes        | <input checked="" type="checkbox"/> | No | <input type="checkbox"/> |
| Handout                 | None       |                                     |    |                          |
| Statistical/ Log Tables | Yes        |                                     |    |                          |
| Graph Paper             | None       |                                     |    |                          |
| Log Graph Paper         | None       |                                     |    |                          |
| Other Materials         | Calculator |                                     |    |                          |

**Section A** - Answer all parts (a)-(d). (Each part is worth **5 Marks**.)

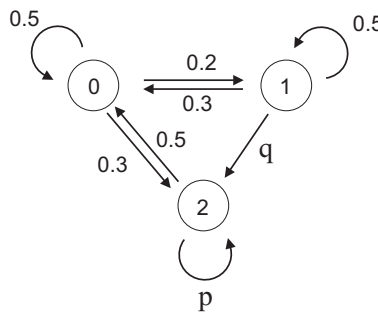
1. (a) Consider a queue in a shop and let

$$f(t) = \begin{cases} \frac{1}{b-a}, & a \leq t \leq b, \\ 0, & t < a, t > b, \end{cases}$$

be the probability density function for a uniform distribution of the waiting time  $t$  before a customer is being served, for  $0 < a \leq t \leq b$ .

- (i) What is the average waiting time?
- (ii) What is the variance and the standard deviation of the waiting time?

- (b) Consider the following Markov Process with transition probabilities shown.



- (i) Find the transition matrix  $P$  and the probabilities  $p$  and  $q$ .
- (ii) Is the Markov process ergodic? Explain your answer.
- (iii) Does the Markov process have unique equilibrium probabilities? Explain your answer. If the Markov process has unique equilibrium probabilities calculate them.

**PTO**

(c) Consider a model for a non-fatal disease in a population  $P$  given by

$$\begin{aligned}\frac{dS}{dt} &= -\alpha SI, \quad S(0) = S_0, \quad \alpha > 0, \\ \frac{dR}{dt} &= \beta I, \quad R(0) = 0, \quad \beta > 0,\end{aligned}$$

where  $S$ ,  $I$  and  $R$  respectively correspond to susceptible, infected and recovered groups of the population. Let  $\rho = \beta/\alpha$  and show that the relation between the groups  $S$  and  $R$  can be described by

$$\begin{aligned}S(R) &= S_0 e^{-\frac{R}{\rho}}, \\ R(S) &= -\rho \ln \left( \frac{S}{S_0} \right).\end{aligned}$$

(d) Consider the linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= x_2, \\ \frac{dx_2}{dt} &= bx_1 + x_2 + u, \\ y &= x_1 + x_2,\end{aligned}$$

where  $x_1 = x_1(t) \in \mathbb{R}$  and  $x_2 = x_2(t) \in \mathbb{R}$  represent the state,  $u = u(x_1, x_2) \in \mathbb{R}$  is the input,  $y \in \mathbb{R}$  is the output and  $b \in \mathbb{R}$  is the control parameter.

- (i) Is this system observable? Explain your answer.
- (ii) Is this system reachable? Explain your answer.

**Section B - Answer any two questions.**

2. Assume that the size of a queue is given by a nearest-neighbour Markov model and that it has a Poisson arrival pattern with parameter  $\alpha_k$  and a Poisson servicing pattern with parameter  $\beta_k$ , where  $k$  is the queue size.

- (a) Let the maximum size of the queue be  $r$ .  
(i) Explain why  $\alpha_r = \beta_0 = 0$ .  
(ii) Assuming that the Markov Process is ergodic, show that the equilibrium probabilities are given by

$$\pi_k = \rho_k \rho_{k-1} \cdots \rho_1 \pi_0$$

for  $0 < k \leq r$  where  $\rho_k = \alpha_{k-1}/\beta_k$ .

**8 Marks**

- (b) Let the queue in (a) have

$$\alpha_k = \alpha, \quad \beta_k = k\beta,$$

where  $\alpha > 0$  and  $\beta > 0$ .

Show that the equilibrium probabilities  $\pi_k$ ,  $k = 0, 1, 2, \dots, r$ , are given by

$$\pi_k = \frac{\rho^k}{k!} \frac{1}{1 + \rho + \frac{\rho^2}{2!} + \cdots + \frac{\rho^r}{r!}}$$

where  $\rho = \alpha/\beta$ .

**6 Marks**

- (c) Let the queue in (b) correspond to a phone exchange with 3 operators. If 30 people call the exchange per hour and each phone call is serviced in 4 minutes, on average. What is the probability that the exchange is full, i.e. if each operator is engaged?

**6 Marks**

3. Consider the nonlinear population model for a population  $P$  given by

$$P_{n+1} = R(P_n)P_n,$$

where

$$R(P) = \frac{rP}{1 + \left(\frac{P}{K}\right)^2}$$

and  $r, K > 0$  are constants.

- (a) Discuss why the function  $R(P)$  is important for determining if the population will grow or decline.

**4 Marks**

- (b) Find the fixed points for this model.

**6 Marks**

- (c) Let  $r = 2$ ,  $K = 10$  and describe the long-term behaviour of the population  $P$  if

- i.  $P_0 = 1$ ,
- ii.  $P_0 = 50$ ,
- iii.  $P_0 = 100$ .

**10 Marks**

4. A two-compartment control system for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u, \quad y = c_1,$$

where  $c_1$  and  $c_2$  are the concentrations of the drug in each compartment,  $u$  determines the flow rate of the drug into compartments 1 and 2 and  $y$  is the concentration of the drug in compartment 2.

(a) Investigate if the system is observable and/or reachable.

**8 Marks**

(b) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where  $k_1, k_2, k_r$  are the control parameters, such that the desired output  $y = r = 4$  and the closed loop eigenvalues for the system are  $\lambda_1 = -2$  and  $\lambda_2 = -1$ .

**12 Marks**